Autonomous Grievance Redressal System for Government Services

Suganya S
Department Of Computer Science and Business Systems
Chennai Institute of Technology
Chennai, India
suganyas.csbs2023@citchennai.net

Kirubakaran N
Department Of Computer Science and Business Systems
Chennai Institute of Technology
Chennai, India
kirubakarann@citchennai.net

Senthil Kumar S
Department Of Computer Science and Business Systems
Chennai Institute of Technology
Chennai, India
senthilkumar@citchennai.net

C Selvaganesan
Department Of Computer Science and Business Systems
Chennai Institute of Technology
Chennai, India
selvaganesanc@citchennai.net

Balaraman Sundarambal
Department Of Computer Science and Business Systems
Chennai Institute of Technology
Chennai, India
sundarambalb@citchennai.net

Senthil Kumar G
Department Of Computer Science and Business Systems
Chennai Institute of Technology
Chennai, India
senthilkumarg@citchennai.net

2025 International Conference on Data Science and Business Systems (ICDSBS) | 979-8-3315-8560-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICDSBS63635.2025.11031909

Abstract— **This research focuses on developing an AI-based autonomous grievance redressal system for government services to address and resolve public issues promptly without any delays. The existing systems for grievance redressal often face several problems like manual prioritization, lack of prioritization of high-priority issues, model not being scalable for all forms of devices, inaccurate categorization of problems based on domains like water, electricity, food, earthquake, etc., and slow response times especially when visual evidence of the affected area is involved. To address these problems this research paper proposes a unique AI-driven grievance redressal model that integrates multimodal inputs, combining text and image data, to overcome these challenges. This model is obtained after a deep analysis of all the existing based on severity. In addition to this, domain-based categorization like water, electricity, food quality, etc., is achieved using Natural Language Processing and Computer Vision techniques. A novel feature of this model is the integration of the option to upload images of the affected areas and process them to enhance ranking based on severity. It also integrates multi-language support to enable wide accessibility. This is achieved by leveraging computer vision techniques and pre-trained models to analyse images based on visual patterns. Overall, this research paper proposes a novel solution to address the existing issues in grievance redressal systems.**

Keywords*— *Grievance redressal, Natural Language Processing (NLP), Computer Vision, Prioritization, Multimodal inputs, Sentimental analysis, Image classification, Deep Learning.

I. INTRODUCTION

The word "grievance" refers to a complaint or dissatisfaction perceived from the injustices or inequities within a system. Globally, these grievance redressal mechanisms have been increasingly recognized as one of the important factors for measuring the quality of E-government services. Every citizen has an absolute right to make a complaint, and it is the responsibility of the government to accept all these complaints and proceed with the necessary action. A government that prioritizes citizens' satisfaction must also prioritize the redressal of their grievances. The government being the biggest service provider must meet people's needs and aspirations on time without any delays, otherwise, citizens will tend to withdraw their loyalty towards them [6]. The grievance redressal mechanism is a system established in almost all digital organizations to receive, record, investigate, redress, analyse, prevent, or take necessary action concerning those grievances filed against them [26]. Effective and timely redressal of public grievances is the hallmark of response and responsible governance, in a democratic country. This has become more important after the implementation of the Right to Information (RTI) Act (2005). The evolution of internet-related services has provided the environment to cherish this important interface in the G2C paradigm [29].

During the evolution of grievance redressal system in India, only a few models existed and those were inflexible for accessing through various types of devices. They were operated manually and no prioritization for high-urgency grievances. Small NGOs use these digitalized grievance redressal systems to address public issues. With the advent of the telecommunication system, helpline numbers were introduced, allowing citizens to lodge complaints remotely. This was the first step toward digitizing the communication system. It reduced the need for physical visits to a certain extent and later this digital revolution led to the development of web portals for grievance redressal. Even though there were limitations like inflexible design, manual prioritization of grievances, etc., Meanwhile the rise of smartphones brought mobile platforms and offered a better user experience with enhanced accessibility and convenience. However, there were no advanced system for issue prioritization, and also had only limited capabilities for processing visual evidence effectively

The Centralized Public Grievance Redress and Monitoring System (CPGRAMS) in India, introduced by the Department of Administrative Reforms and Public Grievance (DARPG) in 2007, is one of the earliest systematic digital initiatives in the latest governance which allows citizens to file grievances online and track their status. This laid the foundation for integrating technology into grievance redressal processes

[20]. These systems were time-consuming because of manual prioritization, lack of transparency, and the absence of an automated tracking process. These challenges eventually brought the need for an advanced autonomous grievance redressal system which provides a faster and more accurate grievance resolution and the capability to handle a large number of complaints. The main motivation behind this research is to revolutionize the grievance redressal process for government services in handling challenges such as manual prioritization, lack of scalability, delayed response, domain-specific categorization, multimodal data input, etc by integrating advanced AI, NLP, and Computer Vision techniques. Therefore, this research paper proposes a novel solution integrating automatic categorization and prioritization of grievances along with the option to upload visual evidence enhancing the effectiveness of grievance redressal systems for the government of India.

II. Literature Review

This research paper is created by analysing various research papers for a better understanding of the existing grievance redressal systems for government services. The paper titled "Automation of Grievance Registration using Transfer Learning" discusses an automated application for grievance redressal using MobileNet V2 architecture enabling efficient classification of grievances for both public and government organizations [1]. Das et al., proposes text mining and sentiment analysis techniques to prioritize and categorize grievances submitted to the respective CM cell into high, medium and low [2]. Divesh et al., developed an Online Social Grievance Redressal System wherein individuals can post their grievances which would then be automatically segregated by using learning algorithm [3]. R.K. Das presents a decision supporting complaint redressal system which uses a sentence-based sentimental analysis technique to enhance the efficiency of the grievance handling mechanism [4]. Mane, S., Chaudhari et al., uses priority algorithm to rank different grievances according to their priority using ML methods [5]. Chaudhary et al., emphasizes that strengthening the public grievance redressal system in Jammu and Kashmir improves accountability, responsiveness and user-friendliness in government services [6]. Shyam P. Joy et al. (2021) proposed a smart grievance redressal model using NLP which prioritizes for immediate administrative action [7]. Marathe et al., in 2016 analysed ICT-enabled grievance redressal systems in Central India, focusing on transparency, technology usage and government relationships [8]. N. Rana et al., investigated about the success of an E-government initiative by validating an integrated IS success model. This study also revealed significant positive contacts between system quality, information quality, and behavioural intention [9]. S. Chander and A. Kush (2012) assessed complaint redressal systems in four Indian states by using particular metrics and presented graphical results to illustrate the findings [10]. The E-Governance framework uses data mining to analyse extensive grievance data, and they also showed systems to enhance transparency and reduce corruption [11], [14], [15], [27]. AI-powered centralized grievance system for efficient complaint resolution, using ML for intelligent analysis and prioritization [19], [25]. Employment of sentiment analysis, and multi-channel access to government services in India has been discussed [19], [21]. GPS-based Complaint Redressed System (GPSCRS) is used to detect complaints from the user's nearest location [24].

III. Existing Grievance Redressal Mechanism Across India

In India, the first common grievance redressal platform, the Centralized Public Grievance Redress and Monitoring System (CPGRAMS) launched by the Government of India under the Department of Administrative Reforms and Public Grievances, serves as a unified platform to address citizen's grievances.

A. Features of CPGRAMS

- Online Accessibility: The platform is available 24x7 for citizens to lodge complaints either online or via a mobile application in integration with the UMANG app.
- Transparency and Tracking: Grievances receive a unique registration ID, enabling users to track the status and appeal if needed.
- Integration: Connects with all central ministries and state governments allowing for a streamlined approach.
- Feedback Mechanism: Complainants can provide feedback on resolutions and appeal for unsatisfactory decisions [28].

B. Limitations of CPGRAMS

- Complaints are processed without automated prioritization which can delay the addressing of critical grievances.
- There is no option for uploading and analysing images, limiting its ability to process visual shreds of evidence.
- Users feel there is inadequate tracking and transparency regarding the actions taken due to the manual prioritization system.
- There is a delay in efficiently handling grievances due to large number of complaints [28].

IV. Proposed Methodology

This research paper focuses on creating an autonomous grievance redressal system by integrating ML, NLP, and Computer Vision techniques for efficient prioritization, classification, severity detection-based images, and resolution of grievances promptly without any delay. The proposed model incorporates features like automated prioritization, domain-based classification, and image uploading features to enable users to include visual evidence, along with an improved feedback mechanism. It uses NLP for text analysis, performs extraction of keywords using advanced algorithms, and sentimental analysis, to automate the grievance prioritization process based on urgency. It also classifies complaints into different domains like water, electricity, food quality, etc., using NLP and utilizes Computer Vision to process the uploaded images to assess the severity of issues. Finally, it incorporates a real-time feedback mechanism where users get notified of grievance status. This innovative approach improves grievance redressal system enabling quick and more accurate responses.

V. Key Features of the Proposed Model

The proposed grievance redressal system mainly focuses on enhancing the efficiency, accuracy, and scalability of the existing grievance management system by combining multiple features such as automated prioritization, classification based on domains, image uploading features for affected areas, multilingual support, and an enhanced feedback mechanism. This model uses NLP, Sentimental analysis, and Computer Vision to improve grievance resolution rate and accuracy.

A. *Automated Grievance Prioritization*

Text analysis with NLP: Utilizes techniques like sentiment analysis, keyword extraction, and text summarization to process the complaint descriptions lodged by users. Enabling an option to escalate grievances if unsatisfied with the resolutions. This enables the users

- Keyword extraction technique: Using NLP the most important keywords like “water”, “damaged roads”, and “electricity outage” are identified.
- Sentimental Analysis: Complaints with urgent, critical language like “urgent”, and “emergency” are assigned with high priority scores and are evaluated for prioritization.
- Scoring based on priority: Based on the keywords, sentiment score, and context, weighted scores are assigned for enabling automated decision on grievances.

B. *Domain-based Grievance Classification*

Categorizing grievances**:** Uses NLP-based classification models to categorize grievances into specific domains (e.g., water, electricity, etc.,).

- Entity recognition: The system identifies entities like locations or type of issue in the text and map it to a specific category.
- Domain Validation: It involves cross-checking with predefined categories to make sure that the problem is routed to the specific department.
- Contextual analysis: It uses context-based embeddings from NLP models like BERT to ensure accurate classification.

The first step involves determining the department in which the particular grievance comes under. To classify grievances department-wise we use a layered hybrid model combining TF-IDF and Logistic Regression. mBERT is used for multi-lingual analysis of the grievances submitted in various regional languages.

C. *Image Upload*

Visual evidence uploading feature enables users to upload visual evidences of the affected areas (e.g., images of flooded areas). It automatically processes these images and analyses them carefully to improve prioritization.

- Severity detection using Computer vision: YOLO (You Only Look Once) is used to detect the severity of the issue based on visual patterns recorded. (e.g., a large pothole) [30].
- Extracts metadata (e.g., GPS coordinates) from pictures to figure out locations.
- Prioritization can be further enhanced based on the severity of the image. (e.g., with critical issues like road accidents). Such complaints receive higher scores.

Reference Models:

ResNet (Residual Networks) extracts the features from images. YOLOv8 supports real-time object detection in pictures [30].

D. *Context-Aware Image Validation (CIV) System:*

The option to upload visual evidence in the proposed model of grievance redressal system improves the model's ability to categorize grievances. But this feature might also include some complexity in case of users uploading irrelevant images which can slow the process flow and detract the system efficiency. To address this problem, this paper proposes the Context-Aware Image Validation framework, which serves as a novel solution combining Artificial Intelligence, NLP, and real-time user interaction to ensure that only relevant images are uploaded and stored.

This feature can be implemented using OpenCV. This tool is chosen for CIV because of its superior performance across key validation metrics like memory efficiency, processing speed and integration capability with multiple devices. Object detection and classification can be employed using YOLOv8 and CLIP. The usage of cloud services like Amazon S3 or Google Cloud Storage can be used for image management. Cosine similarity methods can be used to find the relevance of the uploaded image with the grievance. If the uploaded image shows a high match with the grievance, then it gets accepted. But if it shows only a low match then the model prompts the user for clarification or notifies them to upload a relevant image. It guides users to upload images with a stipulated minimum size to manage storage consumption effectively. This can be achieved by providing clear instructions about the acceptable file sizes on the upload interface.

E. *Multi-Language Support*

Language Flexibility: This model uses tools that support multiple languages, to make it accessible to a wide range of users. Integrating with models like Google Translate API or transformer-based language models like Multilingual BERT (mBERT) enables text understanding and processing among different languages.

F. *Improved Feedback Mechanism*

Instantaneous Updates: This model provides users with real-time updates on the status of their grievances. Citizens who lodge complaints are notified using the push notification system when the grievance is resolved or requires any further action. Once the grievance is resolved, users can rate the handling of the issue and give feedback for improvement.

G. *Algorithm*

- Step 1: Initialize the system and make it ready for user input
- Step 2: Authenticate user credentials and check for required complaint fields like text and image.
- Step 3: Accept user grievances comprising textual data, images, or both.
- Step 4: Extract relevant metadata from the complaint (e.g., time, geolocation).

- Step 5: Use NLP for extraction of keywords and identify priority based on the extracted keywords".
- Step 6: Suppose if an image is attached, perform severity detection using Machine Learning (ML) technique and classify images into appropriate categories.
- Step 7: Save grievance data, metadata, and overall analysis results in the database.
- Step 8: Allocate priority scores using a set of predefined rules.
- Step 9: Grievances are categorized into various domains like water, electricity, infrastructure etc.,
- Step 10: Update the status of the grievance to the users using a notification system.
- Step 11: Enable users to give feedback on the system's response in addition to multi-language supporting feature.
- Step 12: If the grievance is resolved notify the user, spot the grievance as resolved and terminate the process.

  Fig. 1. Shows the workflow of the proposed model.

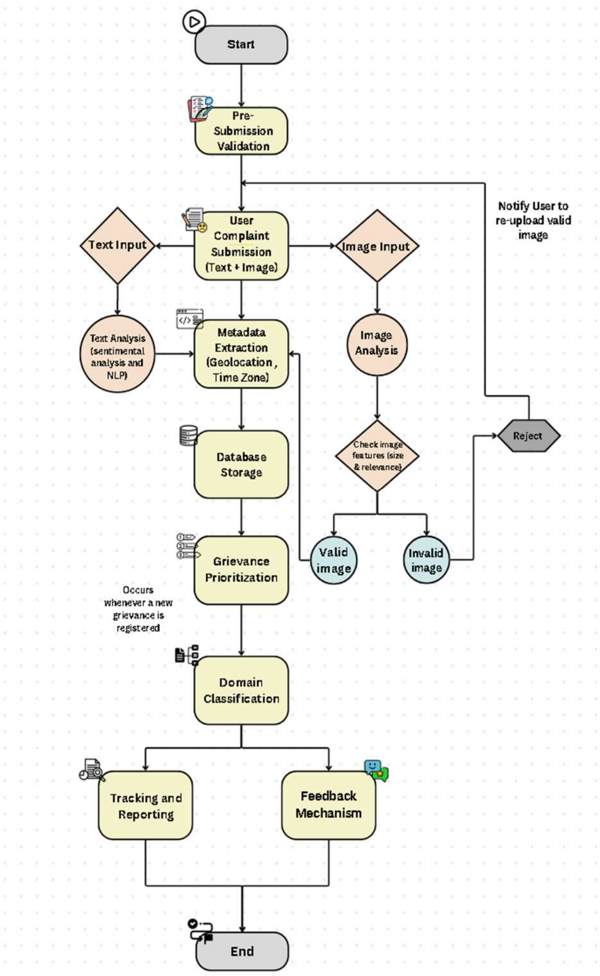

Fig. 1. Workflow Diagram

VI. IMPLEMENTATION

A. Validation of Pre-Submission:

The aim is to ensure that the user inputs (text and image) meet the required criteria. For instance, when you specify that the text input for grievance should have a minimum length of 15 characters, then it means that the users must write a complaint with at least 15 characters in order to submit a valid grievance. This insists users to give enough details in their complaint. For example, if a very short complaint like "Need help" would not be accepted as it is vague and less than 15 characters. Whereas, complaints like "The power supply in our area has been cut off since Tuesday" would be admitted, as it satisfies the minimum character requirement.

Formula:

- For text input:

$$Valid = len(Text) > MinLength \tag{1}$$

- For image input:

$$Valid = (Resolution > MinResolution) \wedge (Size < MaxSize) \tag{2}$$

B. Extraction of Metadata:

In this model, metadata like location, timestamp and device information can be extracted using python tools like Pillow for image metadata and Datetime for timestamp parsing. This information is important for understanding the context of grievance.

Formula:

Metadata for images are extracted by using:

$$Metadata = \{GPS, Timestamp, Device\} \tag{3}$$

C. Text Analysis (Extraction of Keyword using NLP):

The proposed model uses a unique hybrid approach by integrating TF-IDF (Term Frequency – Inverse Document Frequency) + Logistic Regression with BERT (Bidirectional Encoder Representations from Transformers) for handling more complex texts. This approach offers the best in both speed and accuracy. TF-IDF is used for keyword extraction and Logistic Regression for initial classification of complaints into wide range of domains. This initial step reduces the number of grievances that are need to be processed, allowing the model to handle a large-scale dataset. After this initial classification, complex grievances, or those that require contextual understanding like complaints involving mixed sentiments, multiple issues can be processed by BERT for fine-grained analysis. So, in this way TF-IDF, feed ambiguous complaints into BERT for contextual classification. Finally, BERT improves the overall accuracy of categorization.

Formula:

$$TF - IDF(t, d, D) = TF(t, d) \cdot IDF(t, D) \tag{4}$$

where,

- TF: Term frequency in document d.
- IDF: Inverse document frequency across all documents D.
- TF (t, d): Term frequency calculates how frequent a term occurs.
- IDF (t, D): Inverse document frequency lowers the weight of the terms that appear frequently across all document

D. Image Processing and Classification

For those grievances involving visual evidences along with texts, the proposed model utilizes a multi-stage validation system combining Context-Aware Image Validation (CIV), YOLOv8 (You Only Look Once, Version 8) along with CLIP (Contrastive Language – Image Pretraining) to match textual complaints with visual evidences. This helps in the prioritization of complaints as well as helps to reject irrelevant images to the complaint. The CIV system, implemented using OpenCV, first performs a pre-validation check on images to ensure technical compliance such as size, resolution, and format. It uses cosine similarity to assess relevance to the textual complaint. If the similarity score is low, users are notified to re-upload a more relevant image. After passing this initial filtering the images are then processed by YOLOv8 to identify the key visual components from the validated image. These analysed images are sent to CLIP to check if the uploaded image matches with the textual complaint and flags critical images. Open CV is used for CIV because of its speed, lightweight design, and wide-ranging computer vision abilities for validating images before heavy deep learning inference. Empirical studies and benchmarks show that it performs 30-50% faster than PIL or Scikit-Image for image manipulations [35]. It serves as an intelligent gatekeeper allowing only valid images to move into the deep learning pipeline (YOLOv8 + CLIP). The YOLOv8 model for object detection achieves a mean Average Precision (mAP) of ~53.9%, which is a performance metric used to assess the accuracy of detection and localization of objects within images.

- YOLOv8 for deep learning technique implementation.
- It uses a pre-trained model like Darknet-53 as its backbone, which is pre-trained on ImageNet and is highly efficient for real-time object detection
- YOLOv8 classification models are pre-trained to classify objects into 1000 classes, with top-1 accuracy of ~79% for the YOLOv8x model
- YOLOv8x (Extra Large) is the most accurate version (~53.9% mAP) with the largest model size [34].

TABLE I. COMPARISON BETWEEN PROPOSED TOOL AND OTHER TOOLS [34]

Tool	Speed	Accuracy	Best For
YOLOv8	Fastest	High	Real-time applications with large data
EfficientDet	Moderate	Very High	Applications needing precision
Faster R-CNN	Slower	Highest	Complex & cluttered pictures
SSD	Fast	Moderate	Balanced performance for general tasks

E. Grievance Prioritization Process

The model gets inputs from the previous processes and submits those features into the pre-trained XGBoost model. XGBoost assigns priority scores along with flags like high, medium, and low. This model efficiently integrates multi-modal features and scales well for large datasets. Fig. 2 shows the graph of prioritization of grievances using this model.

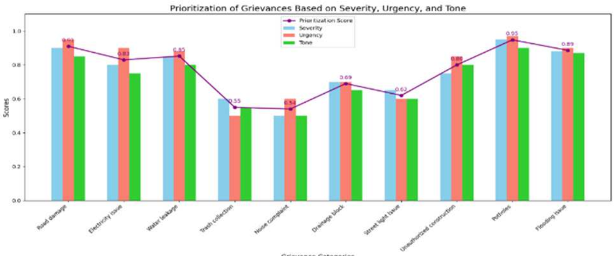

Fig. 2. Graph showing prioritization of grievances

Formula:

$$Priority\ Score = \alpha \cdot Severity + \beta \cdot Urgency \quad (5)$$

F. Database Management

This model employs a multi-layered storage structure to manage different types of data generated during grievance redressal. Raw visual evidence like images and media files, are stored in scalable and secure cloud services like Amazon S3 or Google Cloud Storage. These services offer resilience and high availability for managing large volumes of visual data. Structured data like complaint texts, user details, timestamps, and location metadata are stored in relational database (PostgreSQL). Unstructured data, like image metadata, embeddings, and API responses, are managed in a NoSQL (MongoDB) database. This hybrid approach uses the grievance ID as a common key to link text and image at the time of cross-referencing between the images and the corresponding texts. Using PyMySQL and relevant SDKs/APIS it is possible to create an efficient pipeline that retrieves and processes data dynamically from both databases. This combined approach ensures cost-efficiency and large-scale scalability.

G. Multi-language support

The model proposes the inclusion of a multi-language transformer-based model named Multilingual BERT (mBERT) pre-trained on 100 languages. It is used for understanding different languages and classifying those texts. This tool is ideal for processing complaints that are submitted in regional languages.

Formula:

$$Translation = Encoder(Source\ text) \rightarrow Latent\ Space\ Representation \rightarrow Decoder(Target\ language\ text) \quad (6)$$

Mathematically represented as,

$$T = f_{encoder}(S)\ and\ S_{target} = f_{decoder}(T) \quad (7)$$

where,
S = Source text
T = Translated embedding
S_{target} = output text in the target language.
$f_{encoder}$ and $f_{decoder}$ = Encoder and Decoder functions respectively.

Pseudocode

Input: text_input, img_input (optional), credentials_of_user
Output: Grievance status notification and resolution.
Table II provides the pseudocode of the proposed model.

TABLE II. PSEUDOCODE

```
Initialize system:
    Load pre-trained NLP models (e.g., fastText, BERT)
    Integrate image analysis models (e.g., YOLO)
    Connect to the database
Give authentication to user credentials
if text_input given
    if length(text_input) < MIN_LENGTH then
        return "Invalid input, provide input with valid details"
    end if
end if
if imgtype_input given
    if resolution(imgtype_input) < MIN_RESOLUTION or size_of(imgtype_input) > MAX_IMAGESIZE then
        Return "Image invalid re-upload a valid image."
```

11: end if
12: end if
13: If proper details given accept complaint: text_input, imgtype_input (optional)
14: Extraction of metadata:
Metadata_extracted ← {time, GPS, Device details}
15: Do text analysis:
keywords ← get_keywords(textual_input) (by using TF-IDF)
sentiment_score ← sentiment_analyse(text_input)
16: if imgtype_input given
17: foundsimilar_score ← cosine_similarity (imgtype_input, text_input)
18: if foundsimilar_score < THRESHOLD
19: return "Uploaded image not relevant to complaint."
20: end if
21: found_object ← YOLO.detect(imgtype_input)
22: severity_mark ← computed_severity(found_object)
23: end if
24: Priority score assignment:
prior_score ← α × severity_mark + β × sentiment_score + γ × weight_of_keyword
25: Grievance classification:
category_of_grievance ← fastText.predict(text_input)
26: if category_of_grievance ∉ domains_predefined
27: Return "Needs manual review. Unable to classify."
28: end if
29: Storing grievance data:
Save {text_input, imgtype_input, metadata_extracted, prior_score, category_of_grievance} to the DB.
30: User notification:
Sends push notification with grievance status.
31: Option for user feedback:
feedback_of_user ← fetching_feedback (user_id, grievances_id)
Save feedback_of_user in the DB.
32: Routing grievances to the respective dept:
Update the status to "Grievance in Progress" or "Grievance Resolved" as applicable.
33: if grievance is resolved
34: Notify the corresponding user
35: Mark grievance as "Resolved"
36: end if
37: End the process

VII. Predicted Efficiency Metrics of the Proposed Model

TABLE III. PREDICTED EFFICIENCY METRICS OF THIS MODEL

Process	Accuracy (%)	Time Efficiency (ms)	Success Rate (%)
Input Validation	97	20	98
Metadata extraction	91	25	90
Keyword Extraction using NLP	94	20	95
Image Classification (CNN)	~ 90 (Based on CNN model performance in similar applications)	100 (Standard performance for models like ResNet)	~88 (Standard success rate)
Database Management	97	10	98
Grievance Prioritization	92	20	93
Domain Classification	85 (Based on NLP/ML models in similar types of applications)	30	94
Multilingual Support	85	30	94

Table III, Fig 3 & Fig 4 represent the efficiency & performance metrics of the proposed model.

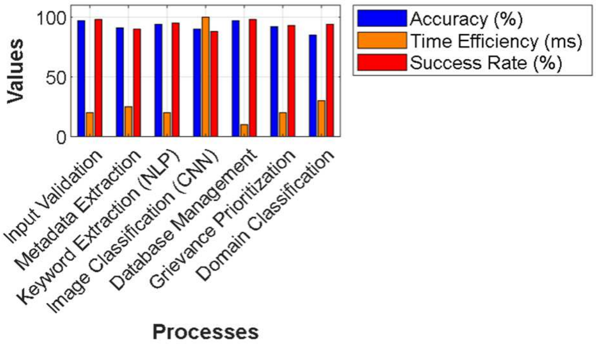

Fig. 3. Performance Metrics of Grievance Handling System

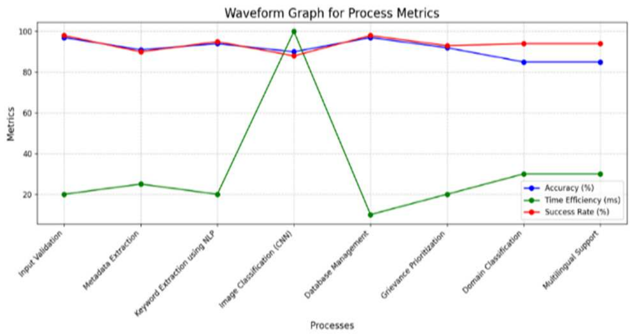

Fig. 4. Performance Metrics in Waveform representation

VIII. Results and Discussions

The proposed autonomous grievance redressal system demonstrates significant improvements over the existing systems, which rely heavily on manual categorizations and prioritization of grievances leading to inefficiencies. In contrast, the proposed model uses an AI-driven automation system for text & image analysis and keyword-based domain classification to ensure fast and accurate categorization and prioritization of grievances. It supports both text & image inputs enabling a much better understanding of the complaint. Present systems follow static rules with manual intervention to determine the complaint priority whereas the proposed model utilizes an AI-based scoring system to dynamically categorize grievances based on the severity. It has an instant real-time feedback mechanism using sentimental analysis, to monitor user satisfaction & improve services. It uses an AI-powered image analysis method to identify and classify physical damages, supplementing textual data for a comprehensive grievance report. It streamlines the grievance redressal mechanism used by our Indian government, from initial submission to resolution, reducing the processing time.

TABLE IV. EFFICIENCY COMPARISON BETWEEN EXISTING AND PROPOSED MODEL

Process	Existing CPGRAMS	Proposed Model	Efficiency Gain
Grievance Classification Accuracy	70% (manual errors)	90-95% (automated by NLP + ML models)	25 – 35%
Prioritization Speed	2-3 hours (17%) (manual)	< 30 mins (83.33%) (automatic scoring system)	66.33% faster
Grievance Resolution Time	7-10 days (75%)	5-7 days (85.71%) due to effective routing)	20 – 30%
User Accessibility	Limited (no ML)	High (multi-language support + feedback)	Vast improvement
Overall Handling Efficiency	75%	90% (automated + structured processes)	15%

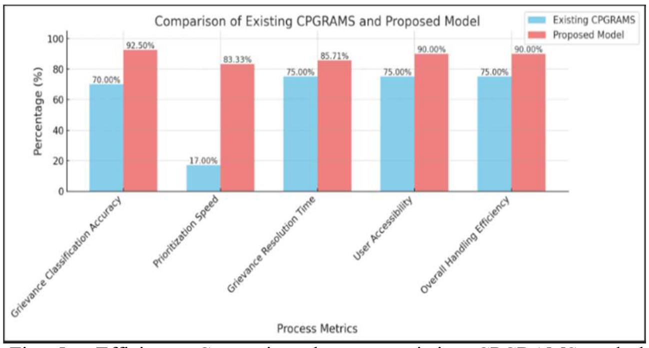

Fig. 5 . Efficiency Comparison between existing CPGRAMS and the Proposed Model

TABLE V. KEY FEATURES COMPARISON BETWEEN PROPOSED MODEL AND EXISTING MODEL

Features	Proposed Model	Indian Govt Systems (CPGRAMS, State Portals, etc.,)
Upload image	Supports image upload with CIV.	Supports basic image uploads no validation.
Processing image	YOLOv8 - object detection & CLIP - image-text matching	No object detection or image-text correlation
Processing Text	TF-IDF & mBERT for understanding complaints	Rule-based/manual processing
Grievance Prioritization	XGBoost for automatic prioritization	Manual processing & no urgency detection.
Multi-Lingual Support	mBERT for multilingual understanding	Limited multilingual support; mostly English + Hindi
Database	Hybrid: PostgreSQL(structured) + MongoDB(unstructured) + cloud storage services.	Traditional RDBMS only; lacks flexibility for complex data
Autonomous Handling	AI-driven routing & response.	Manual routing; slow feedback.

The usage of automated models for complaints prioritization has been explored well. For example, [1] proves the success of transfer learning for mechanizing grievance organization, while [5] highlights an algorithm for prioritization. However, current executions repeatedly emphasize one particular approach, such as keyword identification or sentiment analysis ([22], [4]). The projected system mixes NLP, image processing, and a scoring system to evaluate grievances based on severity and urgency, providing an all-inclusive and scalable prioritization system. Table IV & Fig 5 shows the efficiency comparison between CPGRAMS and proposed model. Table V gives key features comparison between the existing model and proposed model.

TABLE VI. EFFICIENCY GAIN IN PERCENTAGE

Process	Efficiency Gain
Grievance Classification Accuracy	25 – 35%
Prioritization Speed	66.33% faster
Grievance Resolution Time	20 – 30%
User Accessibility	90%
Overall Handling Efficiency	15%

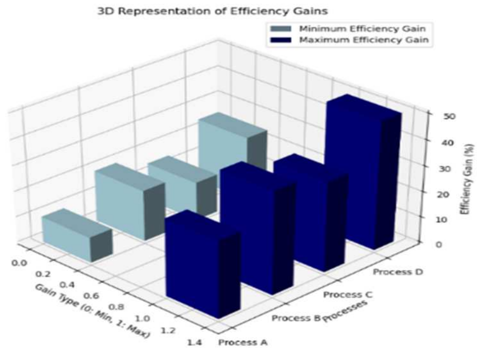

Fig. 6. Efficiency Gains in 3D representation

Text representations highly rely on text-based complaints ([6], [23]) whereas the proposed model exclusively integrates image processing for severity detection and metadata extraction, as proposed by [7], multi-modal inputs advance the effectiveness of problem identification and resolution. While some studies ([9], [20]) emphasize the importance of feedback mechanisms in grievance redressal, existing implementations often rely on static surveys or forms ([18], [15]). The proposed model has a real-time sentiment analysis mechanism, enabling authorities to continuously evaluate public satisfaction and make data-driven improvements. This aligns with findings in [9] that highlight the critical role of feedback for iterative refinement in e-governance systems. Blockchain-based systems, like those explored in [3], ensure transparency and traceability but they highly increase the implementation cost and complexity. Likewise, ICT-based grievance models ([8], [18]) expand convenience but they lack dynamic prioritization techniques and limits multi-modal input capability. The projected model syndicates the recompenses of these methods while extenuating their inefficiencies and limitations. This discussion integrates insights from the existing research and compares how the proposed model addresses the present challenges. Overall, the proposed model designed with AI-based automation reduces dependency on manual intervention, lowering operation costs, improving service efficiency, ensuring seamless scalability and real-time processing, addressing the blockages faced by manual or semi-automated systems. Table VI and Fig 6 shows the percentage of efficiency gain of the given model.

IX. Conclusions

The proposed grievance redressal system effectively addresses the inefficiencies of existing models by implementing innovative features such as AI-driven text and image analysis, dynamic prioritization mechanisms, and multilingual support. This model enhances user experience by enabling accurate and efficient grievance categorization, prioritization, and resolution. It is an advanced model to the existing grievance redressal system in India as it involves automation in prioritization of complaints. Additionally, image processing and metadata extraction provide more understanding of grievances. The real-time feedback mechanism ensures continuous improvement. It lowers operational costs and improves the efficiency of the system by the adoption of advanced technologies, such as AI, NLP and Computer Vision making it an innovative solution for modern grievance redressal system.

References

[1] Sidharth, G., Vasra, A., Sridevi, S., Deisy, C., & Khan, M. (2023). Automation of Grievance Registration using Transfer Learning. *2023 3rd International Conference on Innovative Practices in Technology and Management (ICIPTM)*, 1-6.

[2] Das, R., Panda, M., & Dash, S. (2020). Prioritizing Public Grievance Redressal Using Text Mining and Sentimental Analysis., 273-.284.

[3] Divesh, Manish, Koli., Akash, Bhimani., Drishti, Goda., Kanchan, Dabre. (2020). Online Social Grievance Redressal using Blockchain. Journal of emerging technologies and innovative research, 7(3):215-219-215-219.

[4] Rama, Krushna, Das., Manisha, Panda., Harekrishna, Misra. (2020). 1. Decision support grievance redressal system using sentence sentiment analysis. Doi: 10.1145/3428502.3428505.

[5] Mane, S., Chaudhari, R., Jadhav, V., & Bodakhe, S. (2022). A Multi Factor Algorithmic Approach to Prioritize Grievances. *2022 2nd Asian Conference on Innovation in Technology (ASIANCON)*, 1-6.

[6] Y. udhary, R., & Sugandhar, D. (2020). Strengthening the Institutional Capacities of Public Grievances Redressal System in Jammu and Kashmir: An Analysis. *Indian Journal of Public Administration*, 66, 371 - 379.

[7] Joy, S., Vignesh, N., Shreyas, K., & Roy, A. (2021). Smart Grievance Redressal System. *2021 International Conference on Smart Generation Computing, Communication and Networking (SMART GENCON)*, 1-6.

[8] Marathe,M., O'Neill, J., Pain, P., & Thies, W. (2016). ICT-Enabled Grievance Redressal in Central India: A Comparative Analysis. *Proceedings of the Eighth International Conference on Information and Communication Technologies and Development*.

[9] Rana, N., Dwivedi, Y., Williams, M., & Weerakkody, V. (2015). Investigating success of an e-government initiative: Validation of an integrated IS success model. *Information Systems Frontiers*, 17, 127-142.

[10] Chander, S., & Kush, A. (2012). Assessing Grievances Redressing Mechanism in India. *International Journal of Computer Applications*, 52, 12-19.

[11] Matthey-Prakash, F. (2019). The Grievance Redress System under the Right to Education Act. *The Right to Education in India*.

[12] Meshram, E., Palandurkar, V., Zade, H., Masram, A., & Manmode, N. (2023). A Survey on Student Grievance Redressal System. *International Journal for Research in Applied Science and Engineering Technology*.

[13] Zirtiloglu, H., & Yolum, P. (2008). Ranking semantic information for e-government: complaints management. , 5.

[14] Rana, N., Dwivedi, Y., Williams, M., & Lal, B. (2015). Examining the Success of the Online Public Grievance Redressal Systems: An Extension of the IS Success Model. *Information Systems Management*, 32, 39 - 59.

[15] Rawat, M., & Kaushik, N. (2021). NLP based grievance redressal system for Indian Railways. *ArXiv*, abs/2111.08999.

[16] Rashid Mehmood, Syed Muhammad Amer Mahmood, Muhammad Atif Butt et al. Real-time Co-PPGIS Based Integrated Grievance Redressal Management System Modeling of Public Water Utilities of Lahore, 04 May 2023, PREPRINT (Version 1) available at Research Square [https://doi.org/10.21203/rs.3.rs-2867683/v1]

[17] Chakraborty, D., Ahmad, M., & Seth, A. (2017). Findings from a Civil Society Mediated and Technology Assisted Grievance Redressal Model in Rural India. *Proceedings of the Ninth International Conference on Information and Communication Technologies and Development*.

[18] Divya, Dubey., Dipen, Sahu., Prof.Vikas, Singhal. (2024). Automating Grievance Portal by Automating Central Grievance Redressal System using Artificial Intelligence. International research journal of computer science, 11(02):129-133. Doi: 10.26562/irjcs.2024.v1102.09

[19] Sridhar, Iyer., Deshna, Gandhi., Durga, Prasad, Mishra., Rachana, Trivedi., Tahera, Ansari. (2024). 7. Solve My Problem-Grievance Redressal System. Communications in computer and information science, Doi: 10.1007/978-3-031-50993-3_11

[20] Rao, V. (2015). Citizen Relationship and Grievance Management System (CiR&GMS) through Multi-Channel Access for e-Government Services: A Case from India. *Int. J. E Serv. Mob. Appl.*, 7, 43-67.

[21] Deshmukh, K., & Shiravale, P. (2018). Priority Based Sentiment Analysis for Quick Response to Citizen Complaints. *2018 3rd International Conference for Convergence in Technology (I2CT)*, 1-5.

[22] Bhosale, S., Patankar, S., Kadam, K., Dhere, R., & Desai, M. (2021). Survey on Civil Complaints Management System by Using Machine Learning Techniques. , 606-610.

[23] Badhe, T., Birajdar, M., Mapari, S., & Jorvekar, P. (2016). Grievance Registration on Mobile Application under Smart City Project. *Imperial journal of interdisciplinary research*, 3.

[24] Hani, J., Huzaif, A., & Mohsin, K. (2015). Municipal Corporation Mobile Application for Complaint Resolver. *International Journal for Scientific Research and Development*, 3, 324-327.

[25] United Nations, "E-government survey 2008: From e-government to connected governance," United Nations, New York, 2008.

[26] M. Chouhan, A. Sharma, and R. Jain, "E-governance: Improving transparency and citizen participation in government processes," International Journal of Computer Applications, vol. 52, no. 5, pp. 11–16, Aug. 2012.

[27] Government of India, "Right to Information: A step towards transparency," Ministry of Personnel, Public Grievances, and Pensions, New Delhi, 2012.

[28] Drishti IAS, "Centralized Public Grievances Redress and Monitoring System (CPGRAMS)," [Online]. Available: https://www.drishtiias.com/. [Accessed: Dec. 7, 2024]

[29] Chander, S., & Kush, A. (2012). Assessing Grievances Redressing Mechanism in India. *International Journal of Computer Applications, 52*, 12-19.

[30] L. Jha, S. Miller and J. Wu, "Comparative Analysis of VGG16 and ResNet-50 Architectures for Food Image Classification: A Deep Dive Into Performance and Practical Implications," *2024 8th International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT)*, Ankara, Turkiye, 2024, pp. 1-5, doi: 10.1109/ISMSIT63511.2024.10757277.

[31] Tan, Lu & Huangfu, Tianran & Wu, Liyao & Chen, Wenying. (2021). Comparison of YOLO v3, Faster R-CNN, and SSD for Real-Time Pill Identification. 10.21203/rs.3.rs-668895/v1.

[32] Rapid Innovation. "Object Detection Showdown: YOLO vs Faster R-CNN vs SSD Compared." *Rapid Innovation*, [Online]. Available: https://www.rapidinnovation.io/post/object-detection-showdown-yolo-faster-rcnn-ssd-compared. [Accessed: Dec. 8, 2024

[33] Programmathically. "Deep Learning Architectures for Object Detection: YOLO vs SSD vs R-CNN." *Programmathically*, [Online]. Available: https://programmathically.com/deep-learning-architectures-for-object-detection-yolo-vs-ssd-vs-rcnn/#google_vignette. [Accessed: Dec. 8, 2024]

[34] Keylabs, "YOLOv8 vs Faster R-CNN: A Comparative Analysis," *Keylabs Blog*, [Online]. Available: https://keylabs.ai/blog/yolov8-vs-faster-r-cnn-a-comparative-analysis. [Accessed: Dec. 8, 2024].

[35] ElSayed, A., & Yousef, W. (2019). Matlab vs. OpenCV: A Comparative Study of Different Machine Learning Algorithms. *ArXiv*, abs/1905.01213.